⚡ JavaScript Basics: Make Your Page Come Alive

Variables, decisions, loops, functions and clicks.

Web Dev | Beginner to Intermediate | Ages 10–16

InfiBee Coders

💡 The Big Idea

- JavaScript adds behaviour to web pages: clicks, games, animations and updates without reloading.
- HTML is the structure, CSS is the style and JavaScript is the action.
- You can write it inside `<script>` tags at the end of `<body>`, or in a separate `.js` file.
- `console.log()` prints to the browser console. Press F12 to see it.

📦 Variables and Data

- `let score = 0;` makes a variable that can change. `const name = "Mia";` cannot be changed.
- Types: numbers (5, 3.14), strings ("hi"), booleans (true, false), arrays and objects.
- Join text with `+` or template strings: `` `Hello ${name}` ``.
- Arrays: `let fruits = ["apple", "kiwi"];` and `fruits.length`.
- Objects: `let pet = { name: "Coco", age: 3 };` and `pet.name`.

🔀 Decisions and Loops

- `if (x > 5) { ... } else { ... }` makes decisions. Use `===` to compare.
- `&&` means and, `||` means or, `!` means not.
- `for (let i = 0; i < 5; i++) { ... }` repeats 5 times.
- `while (condition) { ... }` repeats while something is true.
- `for (const x of array) { ... }` goes through every item.

⚙️ Functions and the DOM

- `function add(a, b) { return a + b; }` creates a function.
- The DOM is the page as JavaScript sees it.
- `document.getElementById("id")` finds an element.
- `element.textContent = "Hi";` changes its text.
- `element.addEventListener("click", function(){ ... })` reacts to a click.

💻 Code Examples

Variables and console.log — JavaScript

```javascript
let score = 0;
const player = "Mia";
score = score + 10;
console.log(player + " has " + score + " points");
console.log(`Next level: ${score + 5}`);
```

▶ **Output:**

```
Mia has 10 points
Next level: 15
```

let can change, const cannot. Backticks allow ${...} inside text.

Decisions — JavaScript

```javascript
function grade(mark) {
  if (mark >= 90) {
    return "A";
  } else if (mark >= 75) {
    return "B";
  } else {
    return "Keep going";
  }
}
console.log(grade(95));
console.log(grade(80));
console.log(grade(50));
```

▶ **Output:**

```
A
B
Keep going
```

The function checks the conditions from top to bottom and returns the first match.

Loops JavaScript

```javascript
for (let i = 1; i <= 5; i++) {
  console.log("Star " + i);
}
let count = 3;
while (count > 0) {
  console.log(count);
  count--;
}
console.log("Blast off!");
```

▶ **Output:**

```
Star 1
Star 2
Star 3
Star 4
Star 5
3
2
1
Blast off!
```

The for loop runs 5 times. The while loop counts down.

Arrays and objects JavaScript

```javascript
const fruits = ["apple", "banana", "mango"];
fruits.push("kiwi");
console.log(fruits.length, fruits[0]);
for (const f of fruits) {
  console.log(f.toUpperCase());
}
const pet = { name: "Coco", age: 3 };
pet.age = 4;
console.log(pet.name, pet.age);
```

▶ **Output:**

```
4 apple
APPLE
BANANA
MANGO
KIWI
Coco 4
```

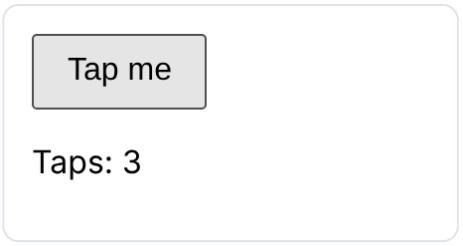
push adds to the end. Objects hold named values.

A click counter — HTML + JavaScript

```
<button id="btn">Tap me</button>
<p>Taps: <span id="out">0</span></p>

let count = 0;
const btn = document.getElementById("btn");
btn.addEventListener("click", function () {
  count = count + 1;
  document.getElementById("out").textContent =
count;
});
```

▶ What you see:



I tapped the button three times, so the page shows 3.

📚 Key Terms

JavaScript
The language that makes web pages interactive.

Variable
A named box that stores a value.

let / const
let can change, const cannot.

Function
A reusable block of code.

Array
An ordered list of values.

Object
A group of named values.

DOM
The page as JavaScript sees it.

Event
Something that happens, like a click.

Event listener
Code that waits for an event.

Console
Where console.log messages appear.

🧮 Quick Reference

```
let x = 5;  const name = "Mia";
console.log(x);
if (x > 3) { } else { }
for (let i = 0; i < 5; i++) { }
function add(a, b) { return a + b; }
const arr = [1, 2, 3];  arr.push(4);
document.getElementById("id")
el.addEventListener("click", fn);
```

✏️ Worked Examples

Q: Show Hello! in a paragraph when a button is clicked.

A: Find the paragraph with getElementById, add a click listener to the button, and set textContent = "Hello!" inside it.

Q: Add up the numbers 1 to 10.

A: let total = 0; for (let i = 1; i <= 10; i++) { total = total + i; } console.log(total) shows 55.

Q: Why does 5 === "5" give false?

A: === compares the type as well as the value. 5 is a number and "5" is a string.

🚫 Common Mistakes

- Using `=` when you meant `===` to compare values.
- Misspelling an id, so `getElementById` returns null.
- Loading the script before the HTML it needs. Put the script at the end of `<body>`.
- Forgetting that array positions start at 0.
- Forgetting semicolons or brackets. The console tells you the line number.

👨‍👩‍👧 Parent Corner

- Ask your child to open their page and press F12 to show you a console.log message.
- Explain: HTML is the bones, CSS is the skin and JavaScript is the muscles.
- Encourage small projects: a button that changes colour, a score counter or a quote generator.
- Ask them to explain one bug they fixed.
- Try free sites like the MDN Web Docs for kids friendly explanations.

🎯 Quick Check

1. What does let do?

✅ It makes a variable whose value can change.

2. What does document.getElementById do?

✅ It finds an element on the page by its id.

3. Which symbol means strictly equal?

✅ === (three equals signs).

4. How do you add an item to an array?

✅ array.push(item).

5. What does addEventListener do?

✅ It runs code when an event, like a click, happens.